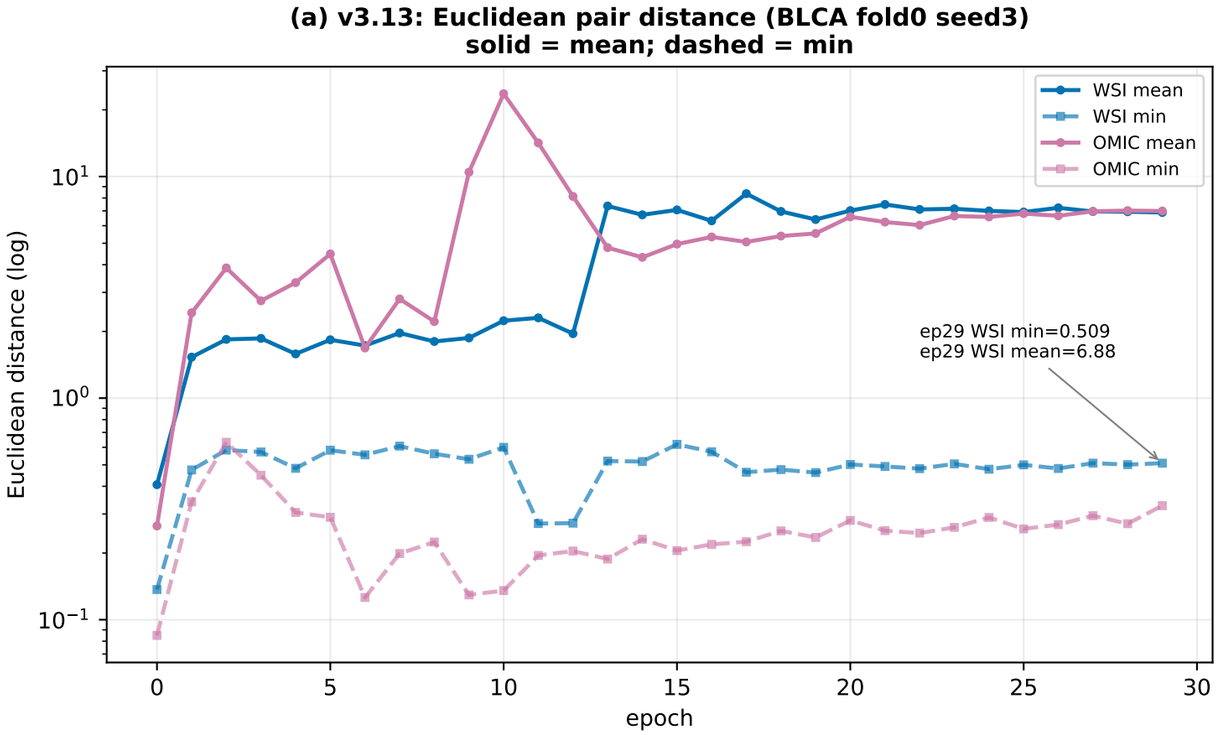
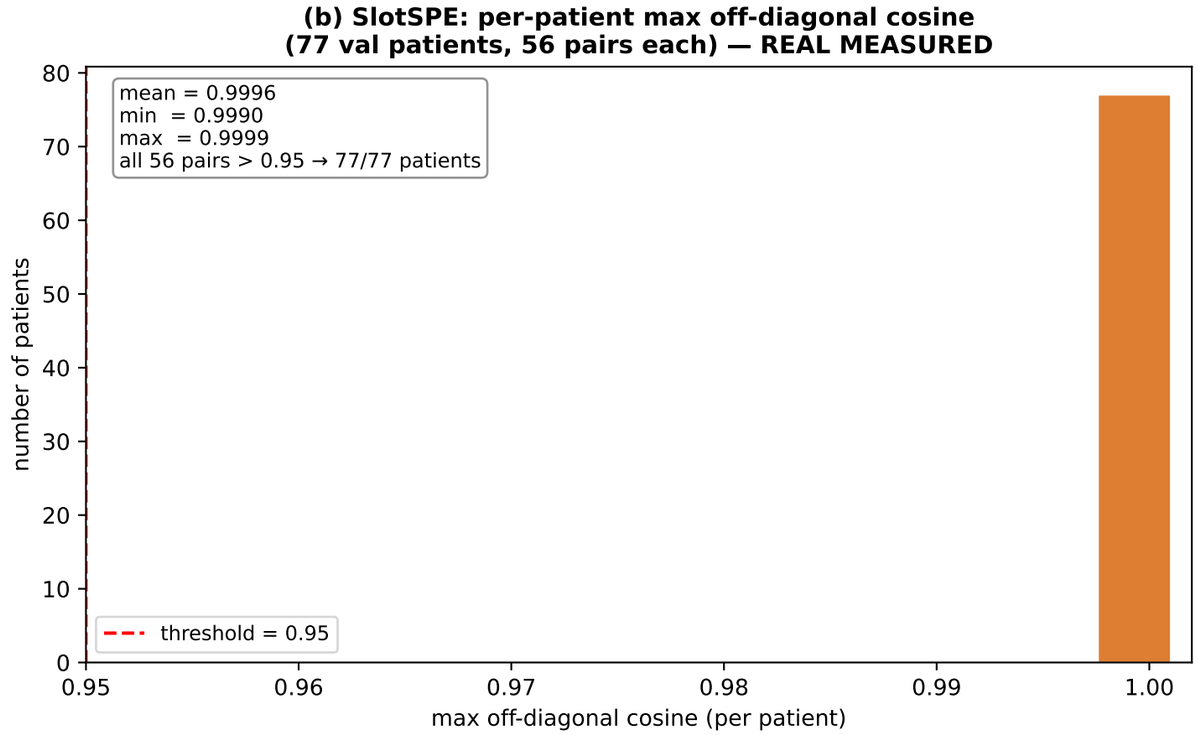
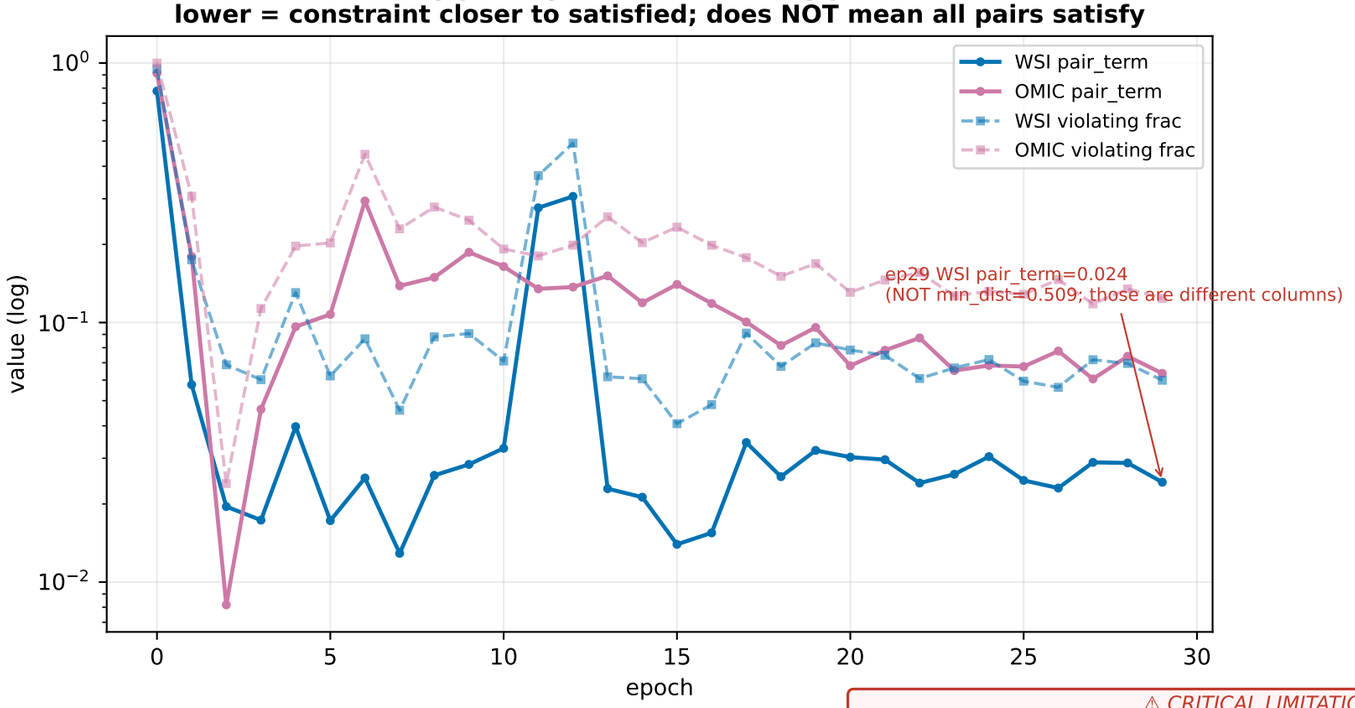
P2 (slot redundancy): v3.13 vs SlotSPE — measured statistics only
⚠ cosine NOT directly measured for v3.13 | ⚠ different protocols not directly comparable







(d) Summary table — all values from real measurements

v3.13 BLCA fold0 seed3 (epoch 29):

WSI	mean pair dist:	6.884
WSI	min pair dist:	0.509 ← measured
WSI	pair_term:	0.0243 (NOT 0.509; see (c))
WSI	violating frac:	0.0599
WSI	duplicate frac:	0.0000 ← pairs with dist=0
WSI	full collapse:	0.0000 ← patients, all 8 identical
OMIC	violating frac:	0.1239
OMIC	full collapse:	0.0000

v3.13 cosine: NOT MEASURED
(estimated lower bound = 0.9997 via $1-d^2/(2r^2)$
with $r \approx 21.87$, equal-norm ASSUMPTION UNVERIFIED)

SlotSPE BLCA fold0 seed3 (epoch 3):

mean	max cosine:	0.9996 ← REAL MEASURED
min	max cosine:	0.9990
max	max cosine:	0.9999
patients all pairs > 0.95:		77/77 ← SINGLE checkpoint
init	max cosine:	0.5605 (FINAL_RECEIPT.md §3)

⚠ CRITICAL LIMITATIONS:
(a–c) are v3.13 only. (b) is SlotSPE only. These use DIFFERENT protocols.
v3.13 cosine: NOT measured. The lower-bound estimate is UNVERIFIED.
SlotSPE data: SINGLE checkpoint (fold0/seed3/epoch3). Not a 5-fold mean.
Do NOT claim 'v3.13 has the same cosine as SlotSPE' or 'v3.13 has lower cosine'.
Do NOT claim 'all 56 v3.13 pairs exceed cos > 0.95' — this is unmeasured.